UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523: DATABASE

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DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN

FOR KOPERASI KADA'S AUTOMATED SYSTEM

Group Name: SYNAPSE

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Stakeholder: Koperasi Lembaga Kemajuan Pertanian Kemubu (KADA) by Mr. Ahmad Rohailan and Ms. Norzamani Ismail

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NOTE: A module can consist of one or multiple (related) use cases. A team member in a group must be responsible for at least one module.

1.0 Introduction

In today's fast-paced digital era, organizations like Koperasi KADA face significant challenges when relying on manual, paper-based processes for managing membership and loan applications. These traditional methods are often error-prone and inefficient. To overcome these limitations, a computerized system is proposed to automate Koperasi KADA's operations. By transitioning to a digital platform, the organization can improve efficiency, reduce errors, and ensure better data management.

1.1 Overview about the company

KADA Corporation is a cooperative managed by Lembaga Koperasi Kemajuan Pertanian Kemubu (KADA), a Malaysian organization dedicated to agricultural advancement. The corporation's main focus is on providing its members with financial assistance such as membership management and loan facilities. With the various range of loan choices customized for different needs, its operations are critical to meet the financial demands of its members.

Currently, the systems at Lembaga Kemajuan Kemubu (KADA) rely heavily on manual processes to manage its koperasi operations, including membership management, loan services and dividend distributions. These operations involve paper-based records and physical documentation, which are not only time-consuming but also prone to human error.

1.2 Problems

KADA Corporation's operations are currently managed through manual processes for membership registration, loan applications, approval notifications, and loan statement preparations. This manual system is time-consuming and prone to errors. KADA corporation is unable to process a large number of applications with the same efficiency due to the lack of an automated system which causes dissatisfaction among members.

There are several issues we found in the current system:

1. There is no automated system for membership registration and loan applications which results in limited applicant capacity for the corporation.
2. Employees must manually inform applicants about membership and loan application approvals by calling them.
3. After approval, employees manually record applicant information which can lead to data inaccuracies because of minor accidents like misspellings and indecipherable handwriting.
4. Employees manually calculate loan monthly repayments and update loan statements for each member.
5. There is a risk of data loss because it is vulnerable to theft, natural disasters, and missing forms due to negligence since all the forms are in paper forms.

1.3 Proposal

To address the challenges faced by KADA, we propose the digitalized of KADA system, which will modernize and streamline its core operations through five modules which are Membership Module, Review Module, Loan Application Module, Approve Application Module and Statement Report Production Module. Such a system will automate manual tasks to minimize errors and enhance overall efficiency for both members and employees of the Koperasi. The modules are as follows ;

1. Membership Module
   The module will let them automate membership registration and approval, applying online and tracking statuses of their applications in real-time.

2. Review Module
   The module is intended to help administrators in membership and loan reviews with the functionality to flag incomplete submissions and record detailed feedback.

3. Loan Application Module
   The module will manage the loan application process by featuring a user-friendly interface with the ability to submit loan requests, upload necessary documents and utilize a built-in loan calculator to estimate the amount of repayments.

4. Approve Application Module
   The module enables the ALK to evaluate applications reviewed by admin against predefined criteria for approval and rejection. This module will prompt ALK to make decision to reject or approve membership and loan application,

5. Statement Report Production Module
   The module produces comprehensive financial reports after the approval or rejection of membership and loan application. This module will also allow admin to send the status of the applications and their statements to the applicants by email.

These five modules will work together to solve the existing operational challenges of the Koperasi and will ensure that the processing is more accurate, transparent and satisfactory for the members. It also focuses on reducing the dependence on paper records and manual labor. The KADA Automated System will go a long way in helping cost savings in the long term, while also enabling the organization to tackle increasing numbers of applications efficiency.

1.4 Definitions*, Acronyms and Abbreviations

This section provides the definitions, acronyms and abbreviations used in the System Documentation (SD)

- ALK : Ahli Lembaga Koperasi - Board members that responsible for decision-making

- Admin : Staff that responsible for managing and reviewing applications
- Applicant : A user that apply membership or loan through the system
- SD : System Documentation - Detailed document

2.0 Database System Requirements

This section describes the essential details needed for designing, implementing and managing the database to support the functionalities of the KADA Automated System. This section will ensure that all data-related requirements related to the system are identified, including data input, updates, queries and structural design. It is divided into several components as follows :

2.1 System Overview

The system is an integrated platform designed to ease the process of membership and loan applications, review as well as approval. This system is also designed for a statement report generation that must be produced after processing membership and loan transactions. It provides a smooth workflow for applicants, administrators and ALK or decision-makers by effectively managing data across multiple modules. Users may apply for membership and loan by providing personal information after inserting their login information and the system also includes tools for tracking and managing application statuses. Additionally, administrators can review, update and finalize membership and loan applications as well as correcting any incomplete submissions as needed. Moreover, the system allows authorized staff (ALK) to approve or reject membership and loan applications based on predetermined criteria and conditions. The system makes it easier to generate and provide thorough financial summaries that conclude customers' loans, savings and other transactions which improves financial management. The system overview can be seen in the Appendix, Figure 4.1 Use Case Diagram.

2.2 Module Membership Application

Safiya Nursyahadah is the person in charge of documenting this module.

2.2.1 Module Description

The membership Application Module includes one use case which is UC01 : Login to Apply for Membership. This module is designed to facilitate applicants in accessing the system to apply for membership. The actor involved in this module is the Applicant, who performs the main function of logging into the system and applying for membership.

Login to Apply for Membership (UC01) :

- The system should allow applicants to login using their credentials to access the membership application form.
- Applicants should provide their login information.
- The system should verify the validity of the provided login credentials.
- If the login is successful, the system should display the membership application form for applicants to fill out and submit.
- If the login credentials are invalid, the system should display an error message and prompt the applicant to re-enter the correct details.

2.2.2 Transaction Requirements: Data Entry

1. Enter the login credentials of the user to access the membership application form.
2. Enter the personal details of the applicant for applying membership.
3. Record login activity.

2.2.3 Transaction Requirements: Data Update/Deletion

1. Update/Delete user login credentials.
2. Update/Delete recorded login details.
3. Update/Delete session information.
4. Update/Delete the applicant's membership application details.

2.2.4 Transaction Requirements: Data Queries

1. List the details of logged-in users who applied for membership.
2. Identify the total number of successful and failed login attempts.
3. List active sessions, including user details and session duration.
4. Identify accounts with repeated failed login attempts.
5. List the membership applications submitted by users.
6. Generate a summary of login analytics.
7. List login history and related session information for a specific user.

2.2.5 Local Conceptual Data Model (ERD) for Module 1

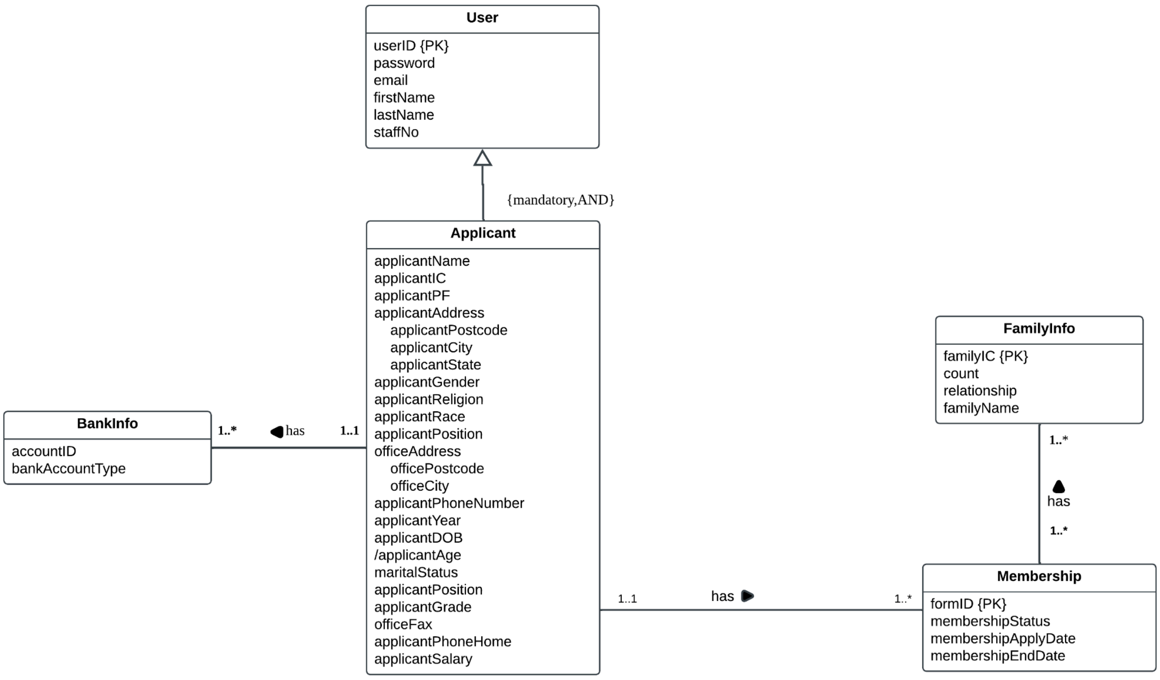

Figure 2.1 Local Conceptual Data Model (ERD) forMembership Application Module

2.3 Module Review

Safiya Nursyahadah is the person in charge of documenting this module.

2.3.1 Module Description

The Review Module includes a use case, UC02 : Review Applicants Form. This module is designed to help administrators efficiently review the applications submitted by applicants for both membership and loan. The actor involved in this module is the Admin.

Review Applicants Form (UC02) :

- The system should allow administrators to access and review submitted membership and loan applications.
- The administrator should be able to view applicant details, including personal information, supporting documents and application status.

- For membership applications, the administrator should verify the submitted information.
- For loan applications, the administrator should review loan details such as loan type, requested amount, and attached supporting document.
- The system should allow the administrator to update the status of applications whether completed or not.
- If necessary the administrator will provide comments to applicants for further clarification for incomplete application and resubmission application.

2.3.2 Transaction Requirements: Data Entry

1. Enter the administrator's login credentials.
2. Enter the criteria for selecting a batch of unreviewed applications.
3. Record the review status for each application in the batch.
4. Enter comments for incomplete applications to specify missing details or issues.
5. Enter the final status of the batch review for reporting.

2.3.3 Transaction Requirements: Data Update/Deletion

1. Update/Delete an application's review status.
2. Update/Delete comments added by the administrator during the review.
3. Update/Delete the list of selected applications in the batch if status changes.
4. Update/Delete incomplete applications flagged for further correction.
5. Update/Delete the batch review summary once the review is finalized.

2.3.4 Transaction Requirements: Data Queries

1. Identify the total number of reviewed and unreviewed pending applications within a batch.
2. Retrieve details of incomplete applications, including flagged issues.
3. Identify the total number of applications marked as "Complete" or "Incomplete".

4. Retrieve a summary report of the batch review, including timestamps and reviewer details.
5. List applications flagged for additional corrective action.
6. Generate a report of batch processing efficiency.

2.3.5 Local Conceptual Data Model (ERD) for Module 2

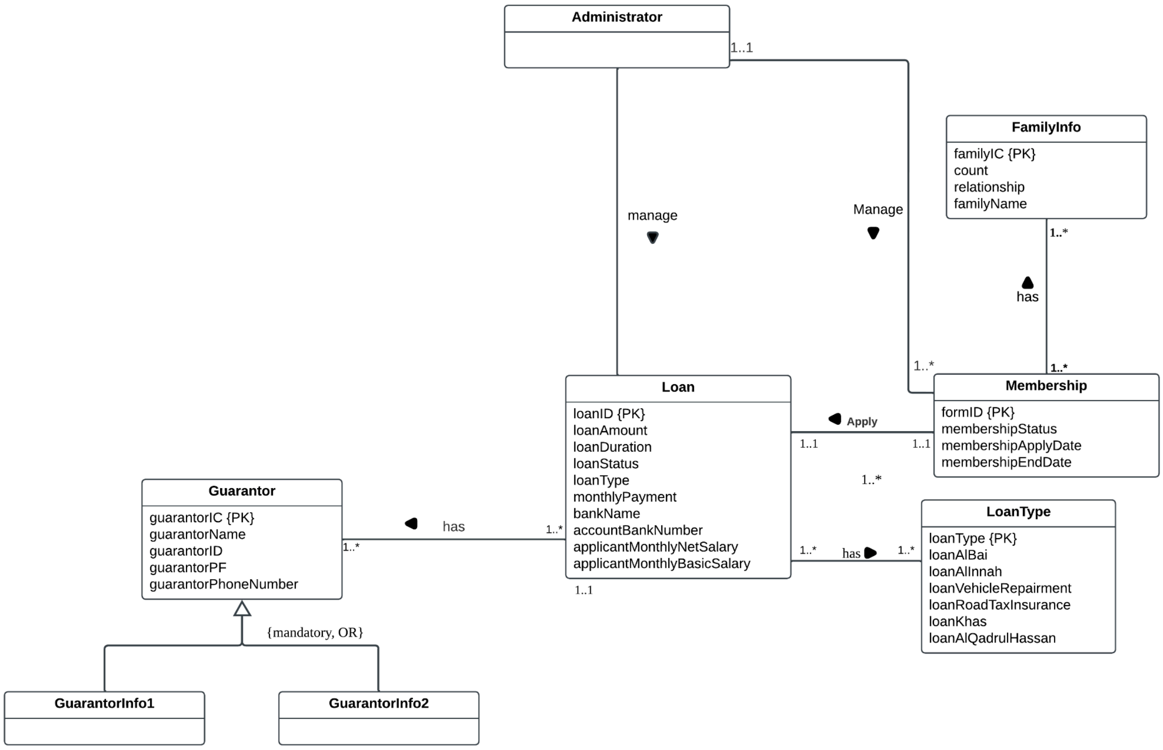

Figure 2.2 Local Conceptual Data Model (ERD) for Review Module

2.4 Module Loan Application

This module is under the responsibility of Farra Nurzahin.

2.4.1 Module Description

The Loan Application Module involves two primary use cases which are UC04: Apply for Loan and UC05: Calculate Loan.

This module is designed to help applicants in completing and submitting loan applications and calculating the estimated monthly loan amount repayment. Applicant, who has the membership and does the main function is the actor involved in this Loan Application Module.

Apply For Loan (UC04):

- The system shall enable applicants to complete and submit a loan application through an online interface.
- Applicants shall provide important information including loan type, amount requests and supporting documents required.
- The system shall automatically validate and verify the given information by the applicants and notify them upon successful submission.

Calculate Loan (UC05):

- The system shall calculate the loan amount based on the applicant's input data.
- The system shall display the estimated loan repayment and monthly installment to the applicants.

2.4.2 Transaction Requirements: Data Entry

UC04: Apply for Loan:

1. Enter the applicant's membership login credentials to access the loan application system.
2. Enter loan details such as loan type and requested loan amount.
3. Upload supporting documents such as guarantor's latest payslip.
4. Enter any additional information such as contact and guarantor details.
5. Submit loan application form for review and approval.

UC05: Calculate Loan:

6. Enter or modify calculation inputs such as required loan amounts.

2.4.3 Transaction Requirements: Data Update/Deletion

UC04: Apply for Loan:

1. Update/Delete loan application details such as loan type, amount requested.
2. Update/Delete supporting documents before submitting the application.
3. Update/Delete incomplete and flagged loan application field for further correction.

UC05: Calculate Loan:

4. Update/Delete the loan amount field input.
5. System Update/Delete the calculation results if error or changes in input entered.

2.4.4 Transaction Requirements: Data Queries

1. Membership detailed to be automatically included in the loan application.

2.4.5 Local Conceptual Data Model (ERD) for Loan Application Module

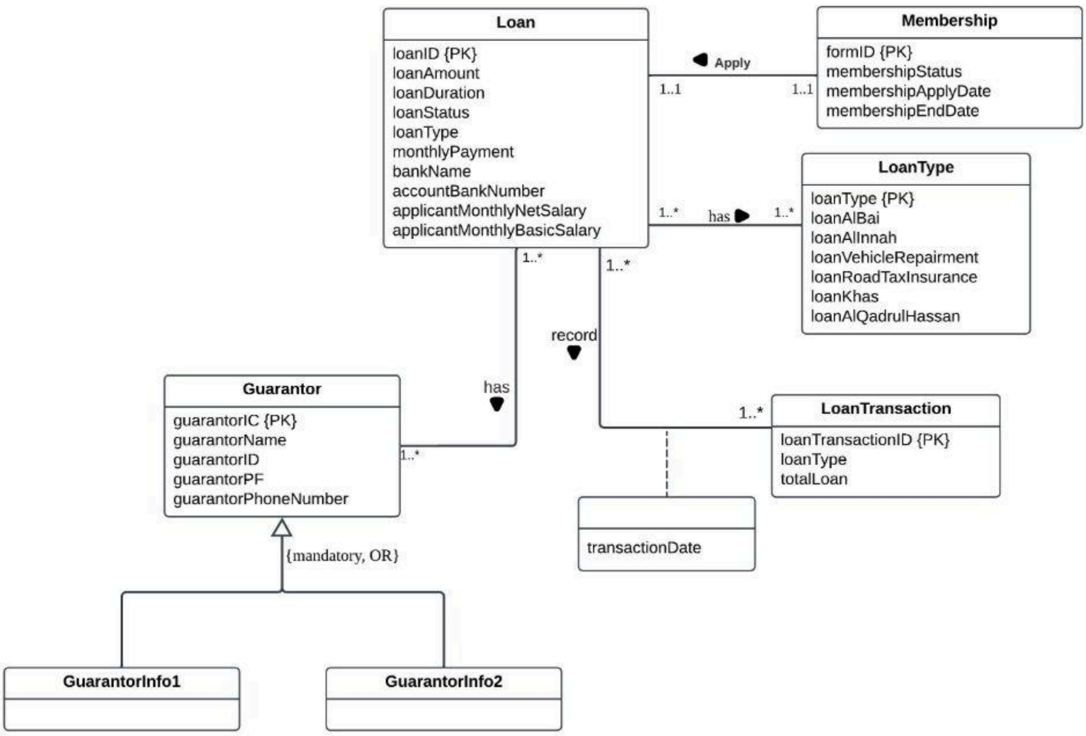

Figure 2.3 Local Conceptual Data Model (ERD) for Loan ApplicationModule

2.5 Module Approve Application

Dayang is responsible for this module

2.5.1 Module Description

This module will involve only ALK. ALK is responsible for membership and loan approval. The use cases inside the module are UC03 : Approve Membership and UC06 : Loan Approval.

Approve Membership (UC03):

- The system shall display the list of pending membership approval
- The system shall allow ALK to approve or reject membership application

Loan Approval (UC06):

- The system shall display the list of pending loan approval
- The system shall allow ALK to compare loan application with loan terms
- The system shall allow ALK to approve or reject loan application

2.5.2 Transaction Requirements: Data Entry

UC03 :

1. Enter the ALK's login credentials to access the membership approval system.

UC06 :

2. Enter the ALK's login credentials to access the loan approval system.

2.5.3 Transaction Requirements: Data Update/Deletion

1. Update/Delete membership status
2. Update/Delete loan status

2.5.4 Transaction Requirements: Data Queries

1. List of pending membership approval
2. List of pending loan approval
3. List of loan terms
4. Identify pending, approved, or rejected membership applications.
5. Identify pending, approved, or rejected loan applications.

2.5.5 Local Conceptual Data Model (ERD) for Module 1

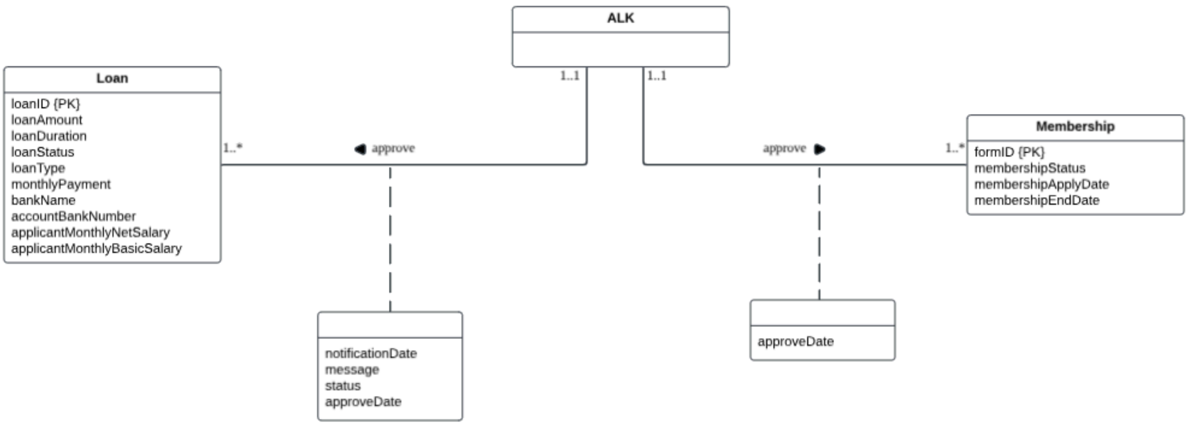

Figure 2.4 Local Conceptual Data Model (ERD) for Module Approval Application

2.6 Module Statement Report Production

Nawwarah Auni is responsible for this module.

2.6.1 Module Description

The Statement Report Transaction Module enables the admin to efficiently prepare and send detailed statements and reports for applicants. This module focuses on generating comprehensive statement reports that summarize the total amounts for savings and loans.

This module includes two key use cases: UC07 (Prepare Statement and Report) and UC08 (Notify via Email). UC07 allows the admin to create and manage statement reports by retrieving the information from the database, while UC08 handles notifying applicants via email about their financial details.

2.6.2 Transaction Requirements: Data Entry

1. Record each transaction whenever an applicant contributes to their savings.
2. Record each transaction whenever an applicant receives a loan.

2.6.3 Transaction Requirements: Data Update/Deletion

1. Allow updates to existing transaction records.
2. Provide the ability to delete incorrect or outdated transactions.

2.6.4 Transaction Requirements: Data Queries

1. Retrieve list of approved membership and loan application.
2. Retrieve a list of all transactions including both savings and loan records.
3. Summary of total savings and loan amounts for each applicant.
4. Fetch detailed information about members such as name, IC number, PF number and staff number.
5. Fetch member’s email address.

2.6.5 Local Conceptual Data Model (ERD) for Statement Report Production Module

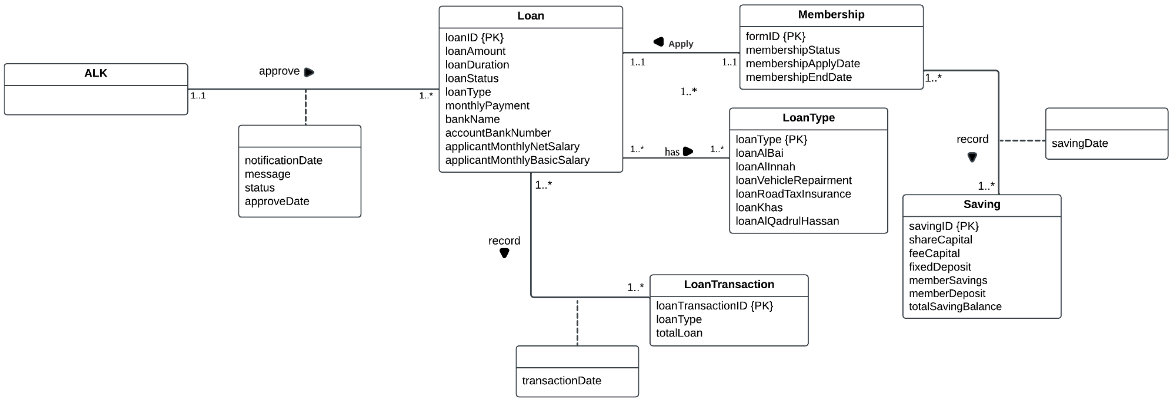

Figure 2.4 Local Conceptual Data Model (ERD) for Module Approval Application

3.0 Global Conceptual Data Model (Global ERD)

This is the ERD as a result of merging the local data models as given in section 2.1 to 2.4 above.

3.1 Global Conceptual Design (ERD)

The Global Conceptual Data Model (ERD) provides an overview of the loan and membership management system, showing relationships between entities. The User entity connects roles such as Applicants, Administrators and ALK. Applicants can apply for Loan and Memberships with loans must be included with the Guarantor information. They can also calculate the estimated monthly loan repayment using the LoanCalculator. Loan details and transactions are stored in Loan and LoanTransaction entities. Membership tracks application statuses and FamilyInfo for family details must be included while applying for the membership. Financial records are managed under Saving, while BankInfo stores banking details. ALK manages the approvals, ensuring smooth interactions across all entities in the system.

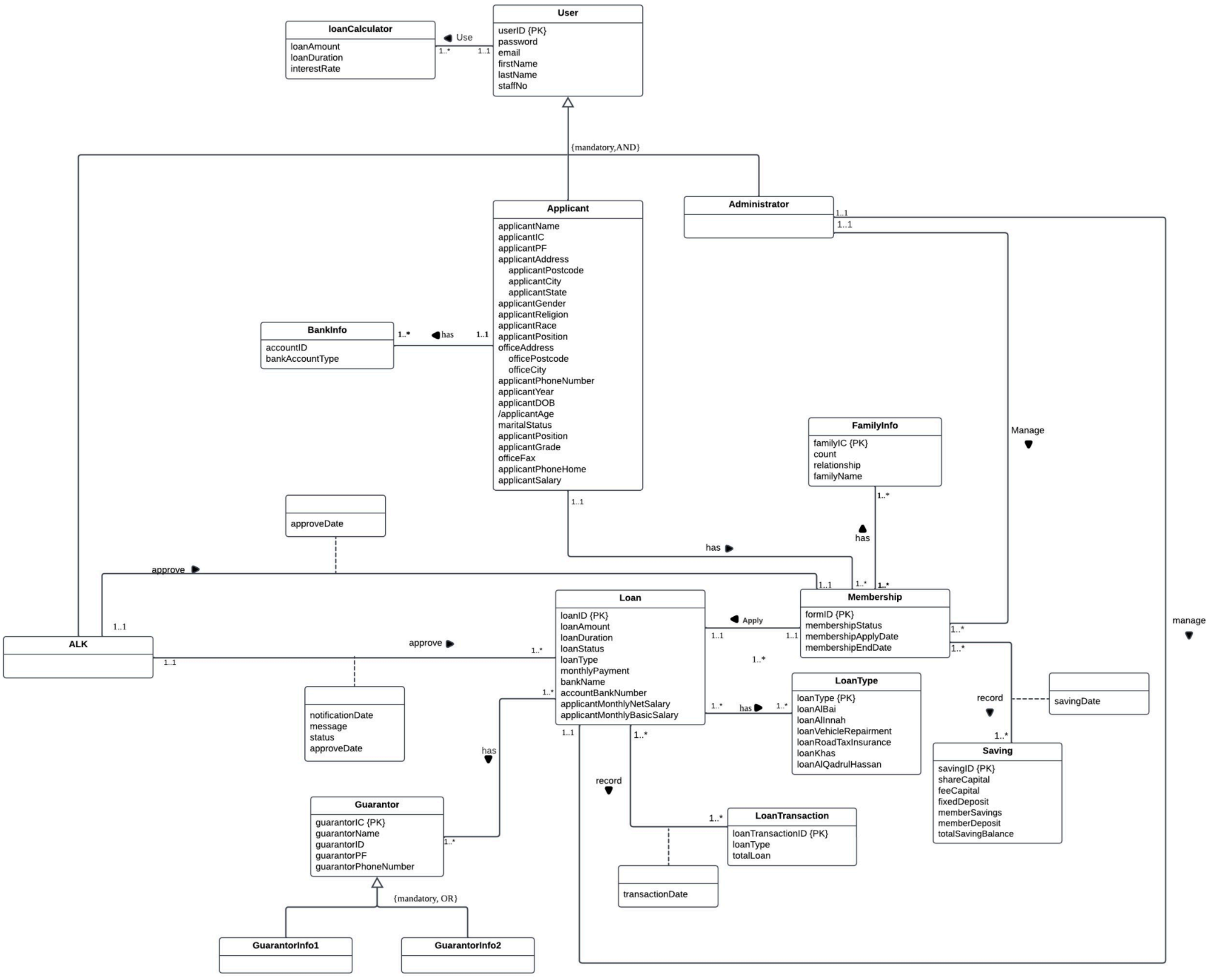

Figure 3.1 Global Conceptual Design (ERD)

3.2 Verified Global Conceptual Design (Verified ERD)

The Verified Global ERD helps organizations manage data more effectively by clearly showing how different parts of the system are connected. It ensures data is accurate, organized, and easy to understand which reduces errors during system design and development. This makes it easier to build reliable databases that meet the needs of the organization. It also improves data consistency, making sure all information stays correct and up to date.

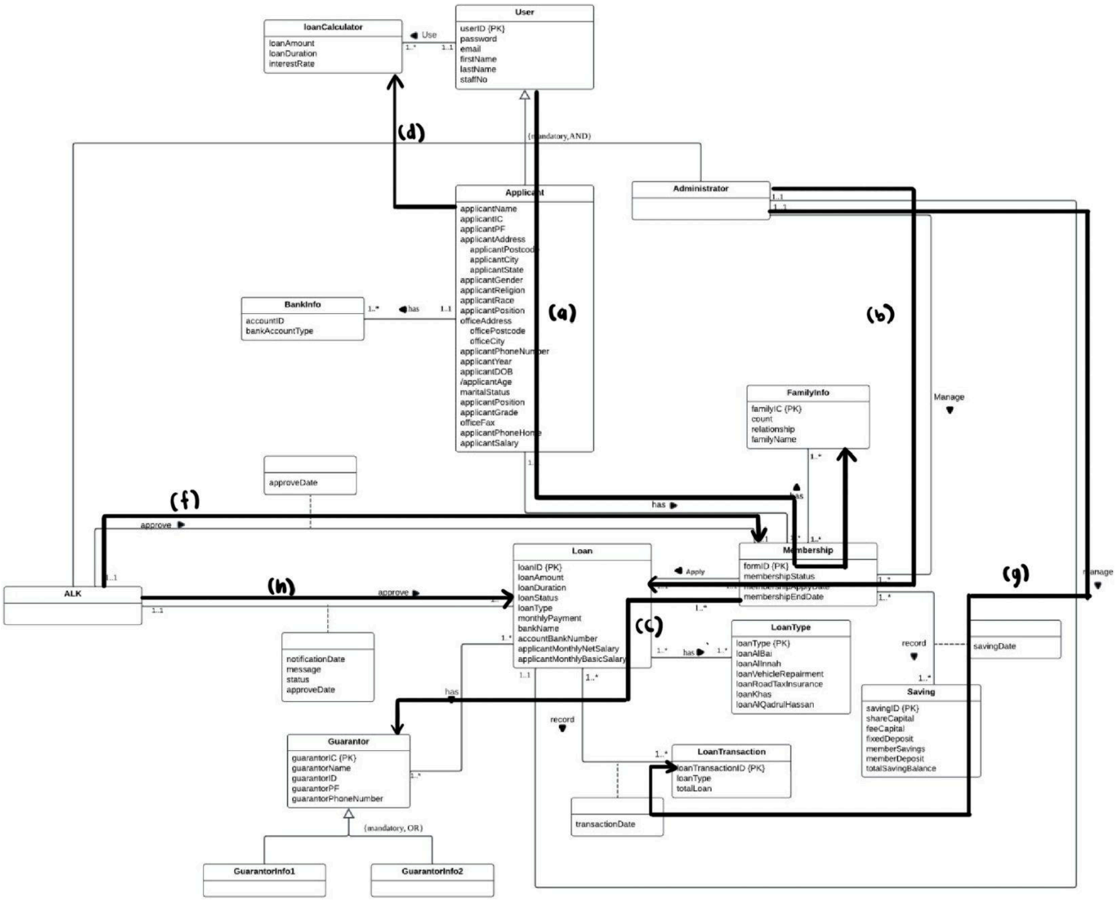

Figure 3.2 Global Conceptual Design (ERD)

The table 3.1 provides a mapping between symbols (a to h) and their corresponding transactions in the Global Conceptual Design (ERD). Each symbol represents a specific part of the ERD, with transactions listed as numbered operations. These transactions highlight the processes associated with the entities and relationships, ensuring clarity in the system's functionality and flow.

Table 3.1 Transaction Global Conceptual Design (ERD)

Symbol	Transaction
a	2.2.2.1, 2.2.2.2, 2.2.3.1, 2.2.3.4, 2.2.4.1, 2.2.4.5
b	2.3.2.1, 2.3.2.2, 2.3.2.3, 2.3.2.4, 2.3.2.5, 2.3.3.1, 2.3.3.2, 2.3.3.3, 2.3.3.4, 2.3.4.1, 2.3.4.2, 2.3.4.3, 2.3.4.5,
c	2.4.2.1, 2.4.2.2, 2.4.2.3, 2.4.2.4, 2.4.2.5, 2.4.3.1, 2.4.3.2, 2.4.3.3
d	2.4.2.6, 2.4.3.4, 2.4.3.5

e	2.5.2.1, 2.5.3.1, 2.5.4.1, 2.5.4.4
f	2.5.2.2, 2.5.3.2, 2.5.4.2, 2.5.4.3, 2.5.4.5
g	2.6.4.1, 2.6.4.2, 2.6.4.3, 2.6.4.4, 2.6.2.1, 2.6.2.2, 2.6.3.1, 2.6.3.2
h	2.6.4.5

3.3 Data Dictionary for the global conceptual ERD

A data dictionary includes important details about the entities, relationships, attributes, and attribute domains within a system. Entities represent objects or concepts, such as a customer or product, while relationships show how these entities are connected. Attributes define the properties of entities, like a staff's name or age. The attribute domain specifies the allowed values for each attribute, ensuring consistency and preventing errors. By clearly documenting these elements, a data dictionary helps maintain a well-structured, understandable system that ensures accurate and consistent data management.

3.3.1 Identify Entity Types

Table 3.2 Data Dictionary Identify Entity Types

Entity Name	Description	Aliases	Occurrence
User	General term describing all the people who used the system.	End user	Each user will access the same system but different activities.
Applicant	General term describing all users logged in as applicants to apply membership.	Membership Applicant	Each applicant will apply for membership after login to the system.
Admin	General term describing the staff responsible for managing and reviewing membership and loan application	System Administrator	Each admin can manage multiple membership and loan applications

ALK	General term describing the authorized personnel responsible for approving membership and loan application	Approving Officer, Decision Maker	Each ALK is associated with membership and loan for approval
Membership	General term describing to the record of applicants who are members of the Koperasi and	Member Record, Applicant Membership	Each membership is only for one applicant
FamilyInfo	General term for the details of the family members associated with an applicant	Family Details, Household Information	Each applicant can provide multiple family member details
BankInfo	General term describing the bank account information provided by applicants for processing membership applications.	Bank Account Details, Bank Information	Each applicant must provide their bank information when applying for membership through the system.
Saving	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.	Savings Record	Each saving record is associated with a single member of Koperasi KADA.
Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.	Loan Application, Credit Request	Each loan is uniquely tied to an applicant and involves submission, review and approval activities within the system.
Guarantor	General term describing a person who agrees to take responsibility for	Financial backer	Each loan application must include at least 2 guarantor details,

	repaying a loan if the applicant has not paid for it.		depending on the loan type and policy requirements.
LoanType	General term describing the different categories or plans of loans offered	Loan Category	Each loan category is essential to each loan application, where the loan application belongs to a member of Koperasi KADA.
LoanTransaction	General term describing the financial activities related to loans, such as disbursements, repayments, and transaction history.	Loan Record	Each loan record is associated with a single member of Koperasi KADA, who can have multiple loan applications.
loanCalculator	General term describing a tool that enables applicants to estimate their monthly loan repayments based on loan amount input.	Repayment Estimator	This feature is accessible to applicants whenever they need to calculate loan repayment estimates prior to submission.
Notification	General term describing a notification update related to an approval process	Notification Log	Each notification is related to membership status and loan status for communication purpose
Approval	General term describing an approval record for tracking the date of approval	Approval Entry	Appears when an application is approved

3.3.2 Identify Relationship Types

Table 3.3 Data Dictionary Identify Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
User	1..1	Use	loanCalculator	1..*
Applicant	1..1 1..1	Has Has	BankInfo Membership	1..* 1..*
Admin	1..1 1..1	Manage Manage	Membership Loan	1..* 1..*
ALK	1..1 1..1	Approve Approve	Membership Loan	1..* 1..*
Membership	1..* 1..1 1..*	Has Apply Record	FamilyInfo Loan Saving	1..* 1..* 1..*
Loan	1..* 1..* 1..*	Has Has Record	Guarantor LoanType LoanTransaction	1..* 1..* 1..*

3.3.3 Identify Attribute and Attribute Domain

Table 3.4 Data Dictionary Identify Attribute and Attribute Domain

Entity Name	Attributes	Description	Data Type and Length	Nulls	Multi-valued

User	userID	Unique identifier assigned for each user	INT AUTO_INCREMENT	No	No
	password	Password for user authentication	VARCHAR(15)	No	Yes
	email	Email for communication and receiving any update information.	VARCHAR(25)	No	Yes
	firstName	Given name of the user	VARCHAR(25)	No	No
	lastName	Family Name of the user	VARCHAR(20)	No	No
	staffNo	Staff Number of the user	INT(10)	No	No
Applicant	applicantName	Full name of applicant	VARCHAR(30)	No	No
	applicantIC	Identification Card number of the applicant	INT(12)	No	No
	applicantPF	Personal File number of the applicant	VARCHAR(30)	No	No
	applicantAddress	Home address of the applicant	VARCHAR(50)	No	No
	applicant Postcode	Postcode of the applicant's address	INT(5)	No	No
	applicant City	City of the applicant's address	VARCHAR(30)	No	No
	applicant	State of the	VARCHAR(	No	No

	State	applicant's address	30)		
	applicantGender	Gender of the applicant	CHAR(1)	No	No
	applicantReligion	Religion of the applicant	VARCHAR(25)	No	No
	applicantRace	Race or ethnicity of the applicant	VARCHAR(25)	No	No
	applicantPosition	Job position of the applicant	VARCHAR(30)	No	No
	officeAddress	Address of the applicant's workplace	VARCHAR(50)	No	No
	officePostcode	Postcode of the applicant's workplace	INT(5)	No	No
	officeCity	City of the applicant's workplace	VARCHAR(30)	No	No
	applicantPhoneNumber	Primary contact number of the applicant	VARCHAR(15)	No	Yes
	applicantDOB	Date of birth of the applicant.	DATE,	No	No
	applicantAge	Current Age of the applicant. This is the derivation of date of birth.	INT(3)	No	No

	maritalStatus	Marital status of the applicant	VARCHAR (10)	No	No
	applicantGrade	Professional grade of the applicant's job position	VARCHAR (10)	No	No
	officeFax	Fax number of the applicant's workplace	VARCHAR (15)	Yes	No
	applicantPhoneHome	Home phone number of the applicant	VARCHAR (15)	Yes	No
	applicantSalary	Monthly Income of the applicant	DECIMAL (10,2)	No	No
Admin	userID	Identifier for admin	INT(10)	No	No
	password	Password for Admin's system login	VARCHAR(50)	No	No
	email	Admin's email address	VARCHAR(50)	No	No
	firstName	Admin's first name	VARCHAR(25)	No	No
	lastName	Admin's last name	VARCHAR(20)	No	No
	staffNo	Admin's staff number	VARCHAR(10)	No	No
ALK	userID	Unique dentifier for ALK	INT(10)	No	No
	password	Password for ALK's ystem login	VARCHAR(50)	No	No

	email	ALK's email address	VARCHAR(50)	No	No
	firstName	ALK's first name	VARCHAR(25)	No	No
	lastName	ALK's last name	VARCHAR(20)	No	No
	staffNo	ALK's staff number	VARCHAR(10)	No	No
Membership	formID	Unique identifier for the membership	INT(10)	No	No
	membershipStatus	Current status of the membership request	VARCHAR(20)	No	No
	membershipApplyDate	Date when the membership starts	DATE	No	No
	membershipEndDate	Date when the membership ends	DATE	No	No
FamilyInfo	familyIC	Family member Identification Number	VARCHAR(15)	No	No
	count	Number of family member	INT(20)	No	No
	relationship	Relationship of applicant with family member	VARCHAR(10)	No	No
	familyName	Name of the family member	VARCHAR(25)	No	No
Notification	notificationDate	Date of notification being sent	DATE	No	No

	message	The message in the notification	VARCHAR(100)	No	Yes
	status	Status of notification	VARCHAR(20)	No	No
	approveDate	Date of approval	DATE	No	No
Approval	approveDate	Date of approval	DATE	No	No
BankInfo	accountID	Account Number for transactions	INT(20)	No	No
	bankAccountType	Name of the bank	VARCHAR(20)	No	No
Saving	savingID	Unique identifier for the saving record	INT AUTO_INCREMENT	No	No
	shareCapital	Share capital amount	DECIMAL (10, 2)	Yes	No
	feeCapital	Fee capital amount	DECIMAL (10, 2)	Yes	No
	fixedDeposit	Fixed deposit amount	DECIMAL (10, 2)	Yes	No
	memberSavings	Member savings amount	DECIMAL (10, 2)	Yes	No
	memberDeposit	Member Deposit amount	DECIMAL (10, 2)	Yes	No
	totalSavingBalance	Total saving balance	DECIMAL (10, 2)	Yes	No
Loan	loanID	Unique identifier for each loan application	INT(20)	No	No
	loanAmount	Loan requested amount	DECIMAL (10, 2)	No	No

	loanDuration	Loan repayment duration	INT(2)	No	No
	loanStatus	Loan application status	INT(10)	No	No
	loanType	Specified loan types	INT(10)	No	Yes
	monthlyPayment	Monthly Payment amount	DECIMAL (10,2)	No	No
	bankName	Bank account name	VARCHAR(20)	No	No
	accountBankNumber	Bank account number	INT(20)	No	No
	applicantMonthlyNetSalary	Monthly income amount	DECIMAL (10,2)	No	No
	applicantMonthlyBasicSalary	Monthly basic salary amount	DECIMAL (10,2)	No	No
Guarantor	guarantorIC	Guarantor identification card	VARCHAR(20)	No	No
	guarantorName	Guarantor full name	VARCHAR(30)	No	No
	guarantorID	Unique identifier for guarantor	INT(20)	No	No
	guarantorPF	Guarantor personal file number	VARCHAR(20)		
	guarantorPhoneNumber	Guarantor phone number	VARCHAR(20)	No	Yes
LoanType	loanType	Unique identifier for the loan type	INT(7)	No	No
	loanAlBai	Loan for Al-Bai	VARCHAR (10)	Yes	No
	loanAlInnah	Loan for Al-Innah	VARCHAR (10)	Yes	No

	loanVehicleRepairment	Loan for vehicle repair expenses	VARCHAR (20)	Yes	No
	loanRoadTaxInsurance	Loan for road tax and insurance payments	VARCHAR (25)	Yes	No
	loanKhas	Special-purpose loan	VARCHAR (10)	Yes	No
	loanAlQadrulHassan	Loan for Al-Qadrul Hassan	VARCHAR (25)	Yes	No
LoanTransaction	loanTransactionID	Unique identifier for the loan transaction	INT AUTO_INCREMENT	No	No
	loanType	Loan Type	VARCHAR (1)	No	No
	totalLoan	Total loan Amount	DECIMAL (10, 2)	Yes	No
LoanCalculator	loanAmount	Loan amount requested	DECIMAL (10, 2)	No	No
	loanDuration	Desired loan duration	INT(2)	No	No
	interestRate	Fixed interest rate for loan	DECIMAL(2, 2)	No	No

4.0 Appendices

4.1 Use Case Diagram for the system

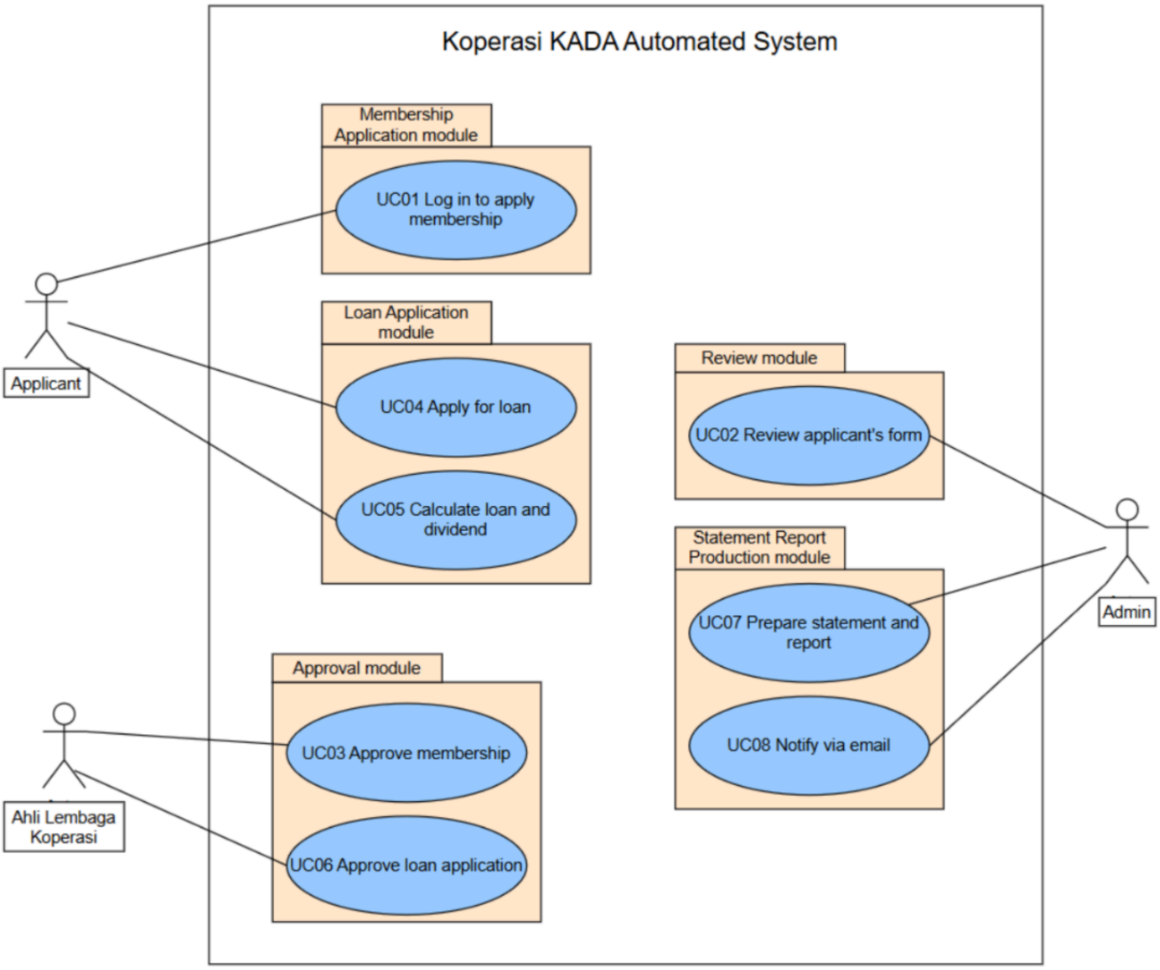

Figure 4.1 Use case diagram

4.2 Activity Diagrams for every use case identified

UC001: Activity Diagram Login for Apply Membership

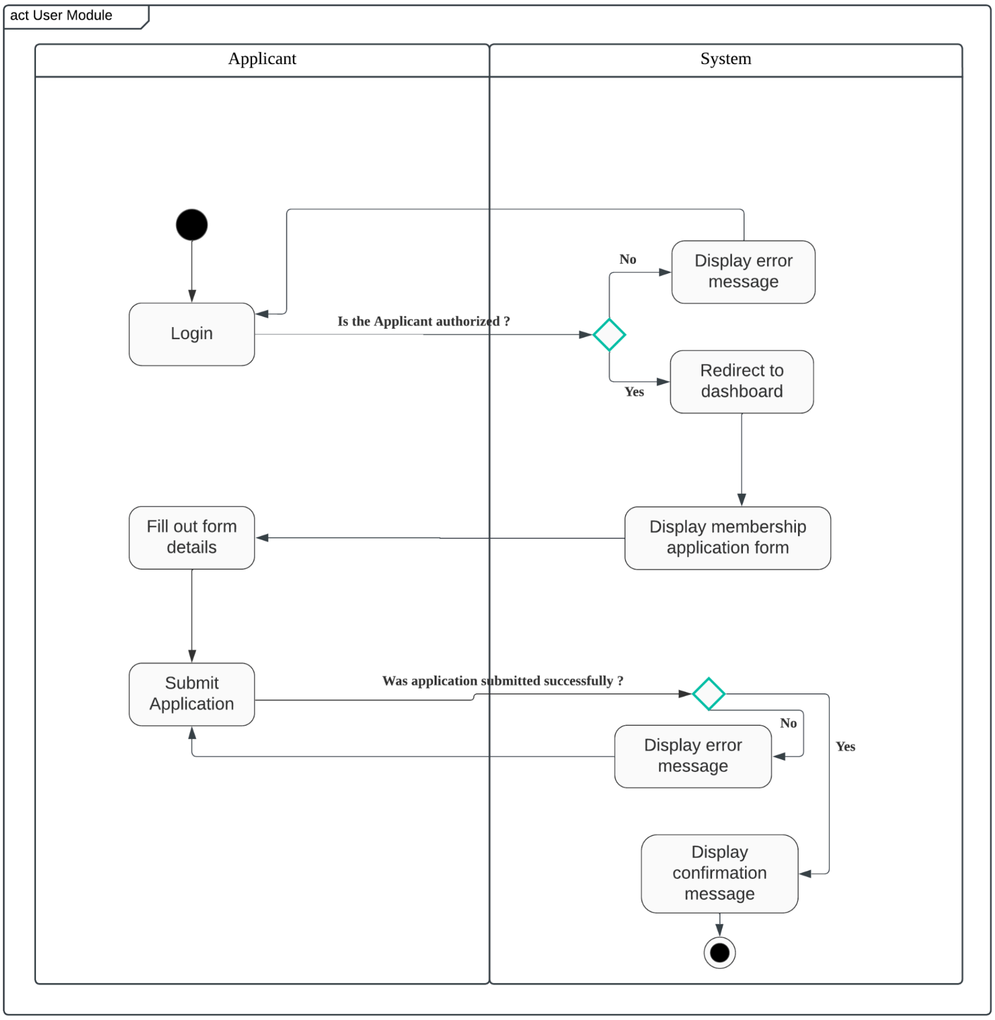

Figure 4.1 Use case diagram for Login for Apply Membership

UC002: Activity Diagram for Review Applicant's Form

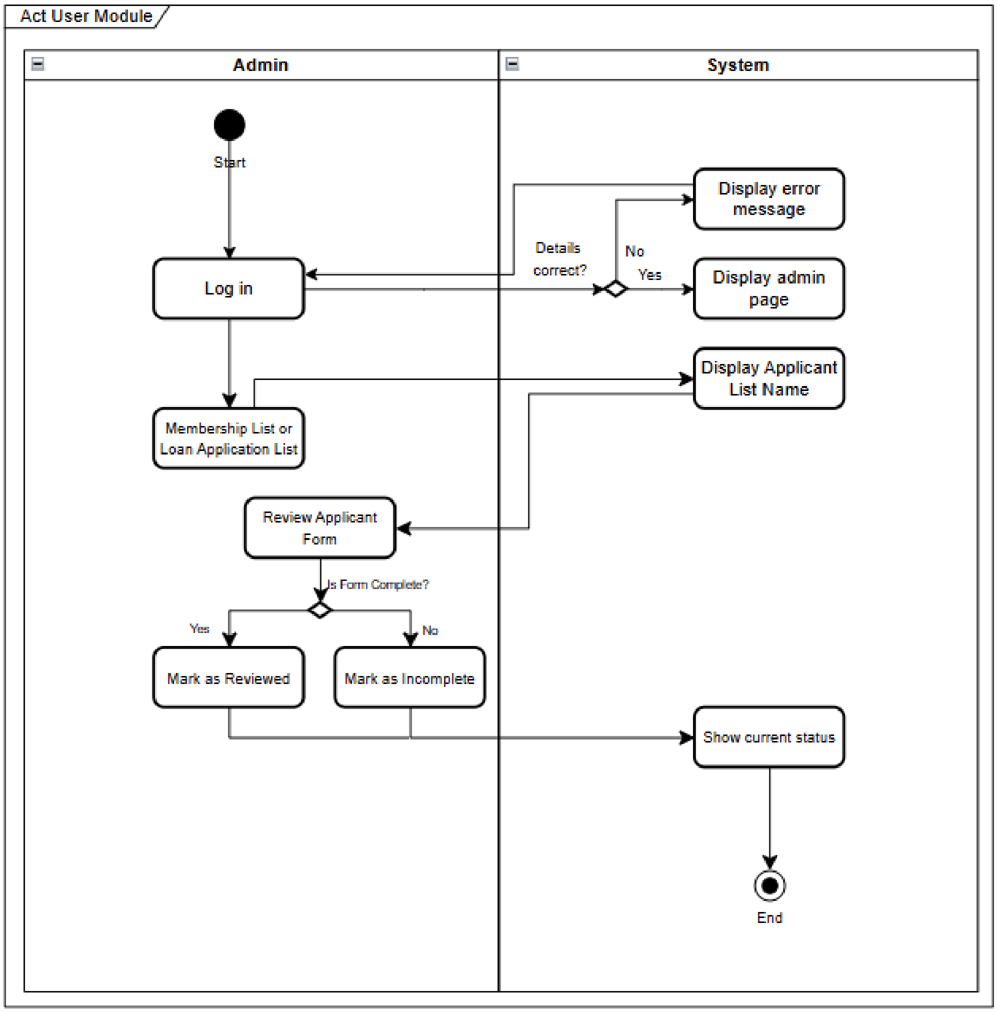

Figure 4.2 Use case diagram Review Applicant's Form

UC003: Activity Diagram Approve Membership

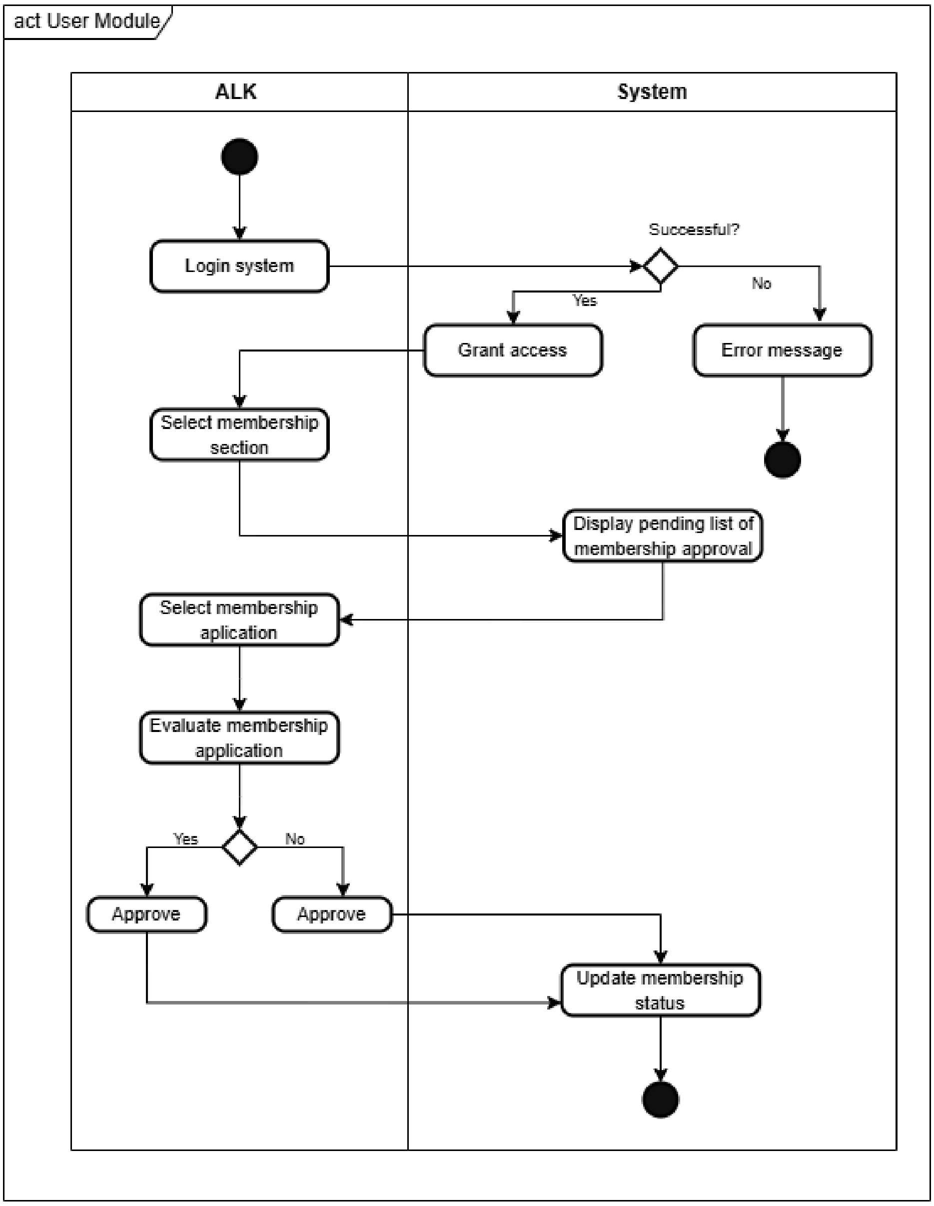

Figure 4.3 Use case diagram Approve Membership

UC004: Activity Diagram Apply for Loan

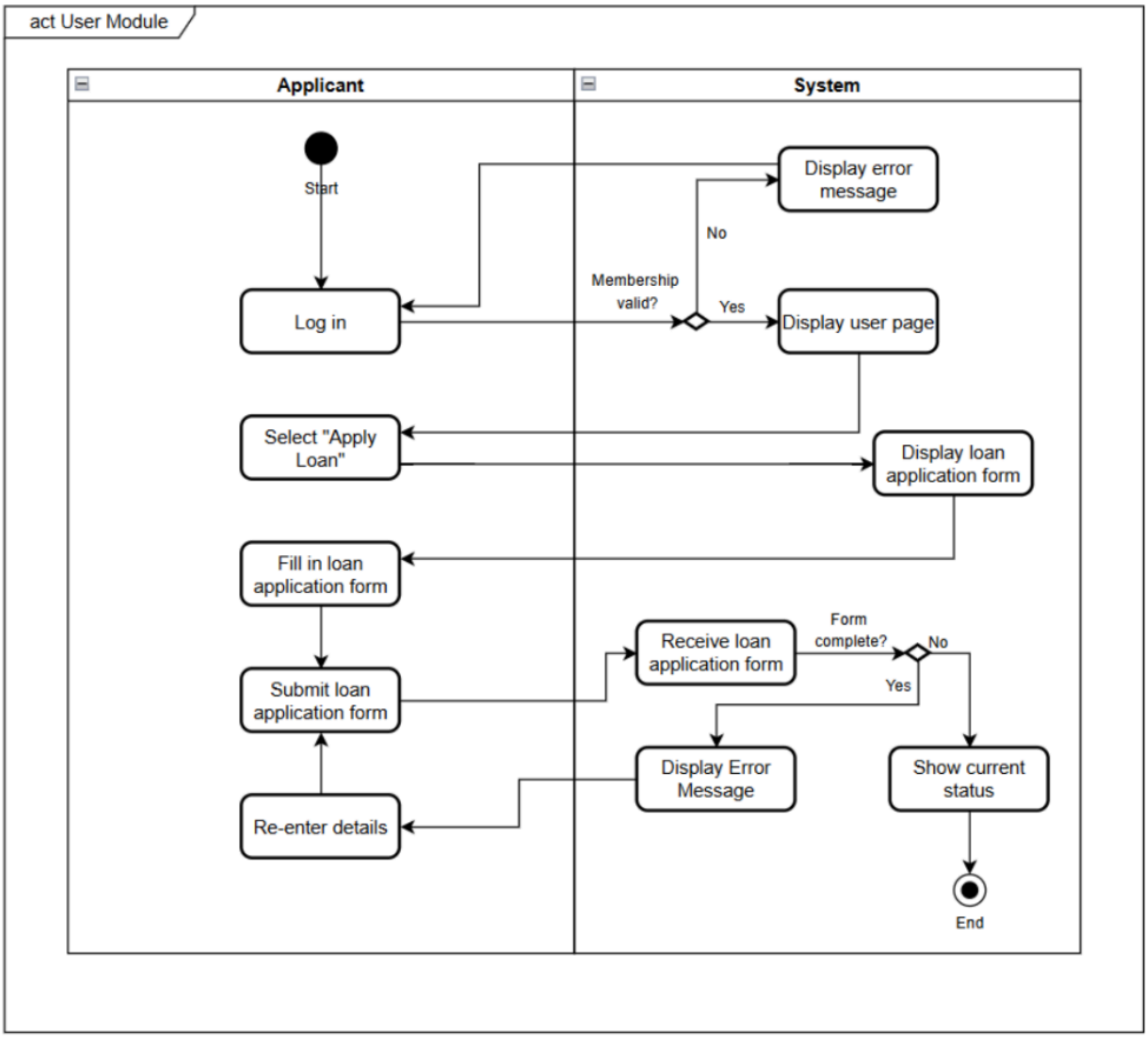

Figure 4.4 Use case diagram Apply for Loan

UC005: Activity Diagram Calculate Loan

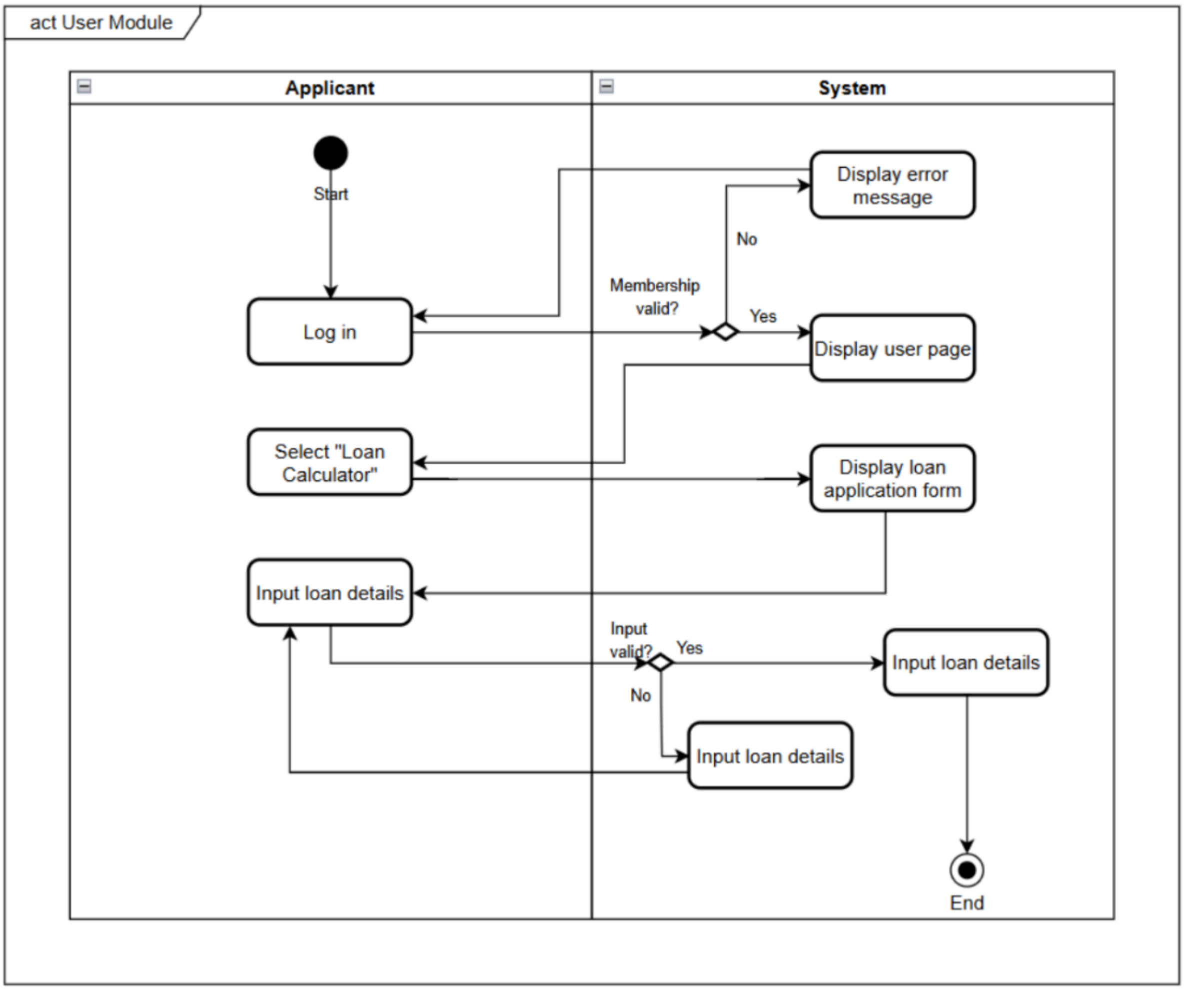

Figure 4.5 Use case diagram Calculate Loan

UC006: Activity Diagram Loan Approval

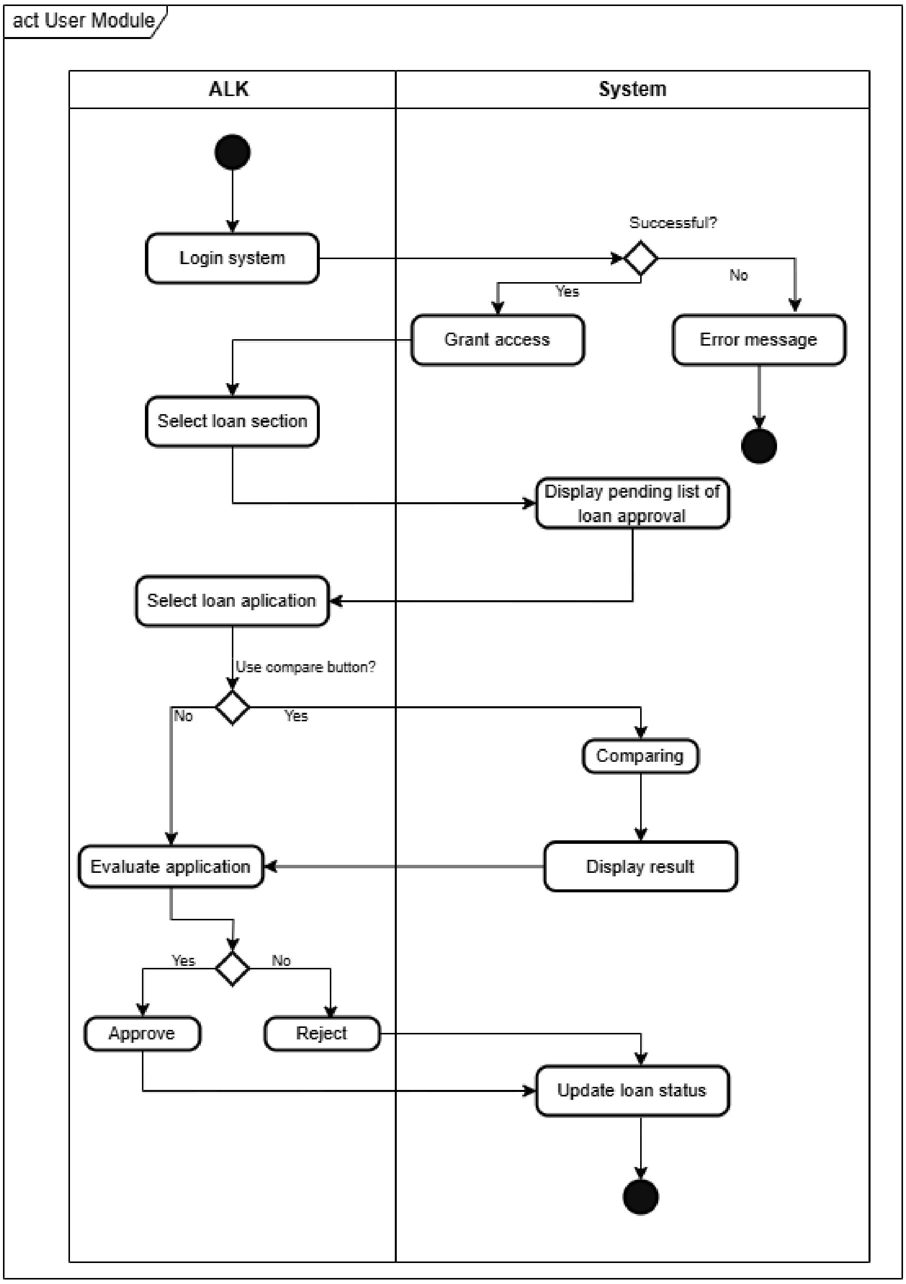

Figure 4.6 Use case diagram Loan Approval

UC007: Activity Diagram for Prepare Statement and Report

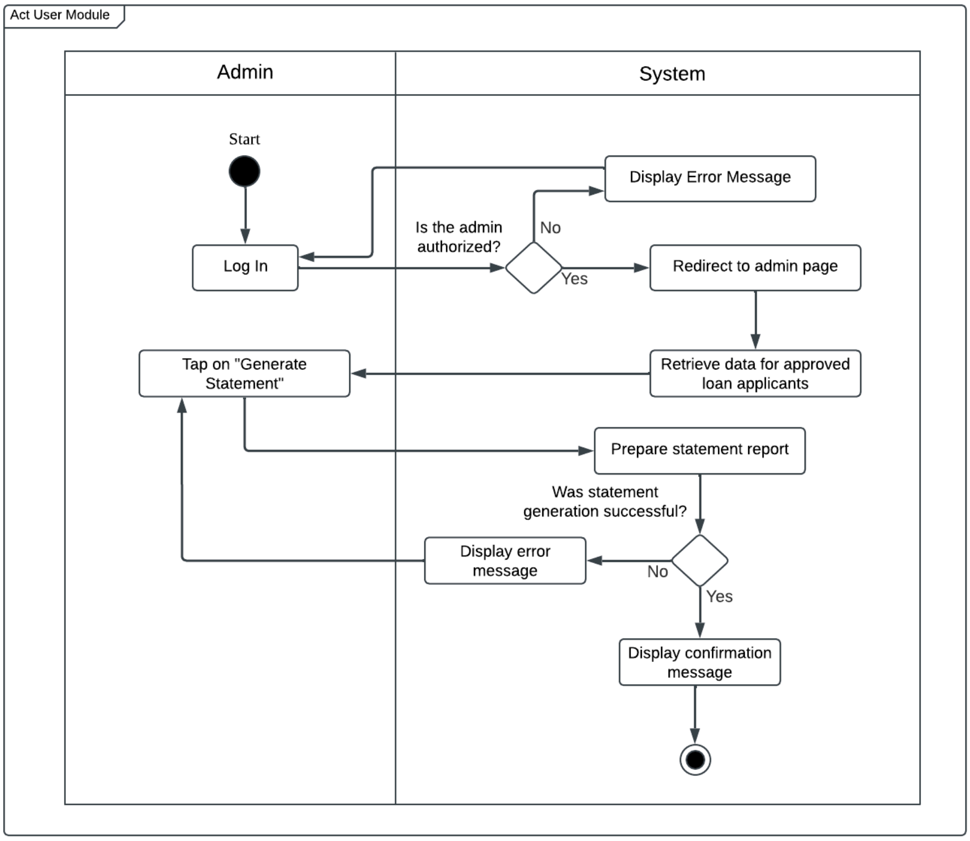

Figure 4.7 Use case diagram Prepare Statement and Report

UC008: Activity Diagram for Notify via Email

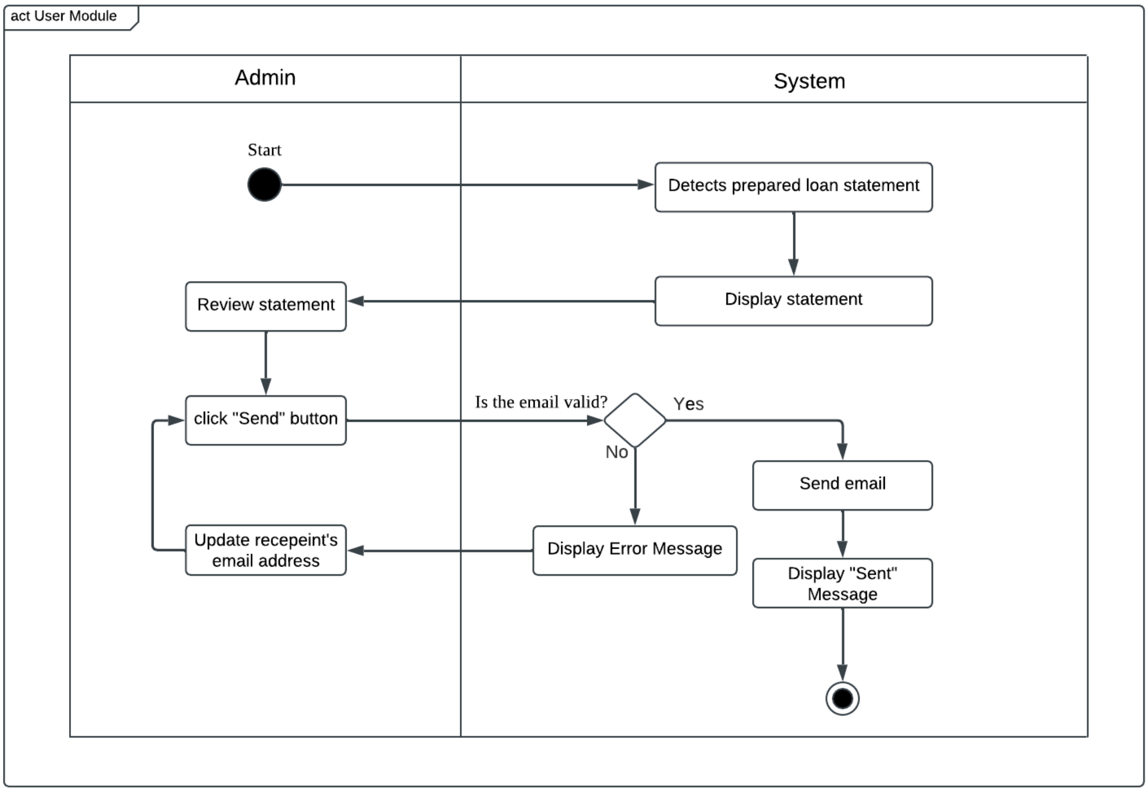

Figure 4.8 Use case diagram Notify via Email

4.3 Meeting Log

Date/Time		11 Dec 2024, 7.30 p.m.
Location		KTDI, M21
Minutes		
No.	Item Discussed	Details
1.	Task Distribution	- All members read the question and rubric together. Then divided the part to each group member
2.	Develop Conceptual ERD	- Each member contributes what entity and attributes should be included in ERD
3.	Explanation of certain part	- Each member contributes to discuss unclear parts of each other

Date/Time		15 Dec 2024, 8.30 p.m.
Location		KTDI, M21
Minutes		
No.	Item Discussed	Details
1.	Do a correction of Conceptual ERD	- Each member contributes to the correction of Conceptual ERD after checking with Dr. Rozilawati.
2.	Do a Verified Global ERD	- All members discussed how to do the Verified Global ERD.
3.	Do Activities Diagram	- Members help each other on activities diagrams.